

# Stage 16N Inhomogeneous Plate-with-Hole Cycle-Jump Benchmark Living Technical Report

Master Thesis Preparation

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## Abstract

This living report records the geometry, numerical setup, HPC configuration, reference results, reinjection tests, diagnostic failures, and current conclusions for Stage 16N. The stage uses an inhomogeneous Abaqus plate-with-hole benchmark with a NEML-equivalent Chaboche UMAT. The full 1000-cycle non-jump reference has been completed and is used as the validation baseline for later fixed and adaptive cycle-jump simulations. The present state of the work is that the global response can be reproduced well after manual SDVINI/SIGINI state injection, but exact local state reproduction near the hole is not achieved by manual state initialization alone. Therefore later cycle-jump errors must be interpreted relative to the measured reinjection error floor.

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## 1 Purpose and Current Status

The purpose of Stage 16N is to move from single-element or homogeneous cycle-jump demonstrations to an inhomogeneous finite-element benchmark. The model is deliberately more demanding: a plate with a central hole produces localized cyclic plasticity near the hole, so the method must be judged not only from global reaction-force loops but also from local stress and state-variable evolution.

Table 1: Current Stage 16N infrastructure and validation status.

| Item                                    | Status                      |
|-----------------------------------------|-----------------------------|
| 1000-cycle full non-jump reference      | Completed                   |
| 16-CPU production policy                | Locked                      |
| Abaqus threaded launcher                | Verified                    |
| Adaptive $\Delta N$ table               | Completed                   |
| Exact reinjection scaffold              | Prepared                    |
| HPC mail policy                         | Fixed                       |
| B0-1 exact reinjection solver run       | Completed                   |
| B0-1 scientific result                  | Practical baseline accepted |
| B0 initialization-only audit            | Completed                   |
| B0-2 / B0-3 exact reinjection jobs      | On hold                     |
| Fixed state-initialized jump validation | Scaffold prepared           |
| Adaptive cycle-jump workflow            | On hold                     |

## 2 Geometry and Model Definition

The Stage 16N model is a three-dimensional plate-with-hole benchmark. It is intended to create localized cyclic evolution near the hole while retaining a global hysteresis response that can be compared across full-reference, exact-reinjection, fixed-jump, and adaptive-jump runs.

Table 2: Stage 16N model geometry and mesh summary.

| Quantity            | Value                         |
|---------------------|-------------------------------|
| Plate length        | 20.0                          |
| Plate height        | 10.0                          |
| Plate thickness     | 1.0                           |
| Central hole radius | 1.0                           |
| Mesh spacing        | $\Delta x = \Delta y = 0.25$  |
| Element type        | C3D8                          |
| Nodes               | 6642                          |
| Elements            | 3148                          |
| Hole-ring elements  | 60                            |
| Material model      | NEML-equivalent Chaboche UMAT |
| State variables     | 27 SDVs                       |

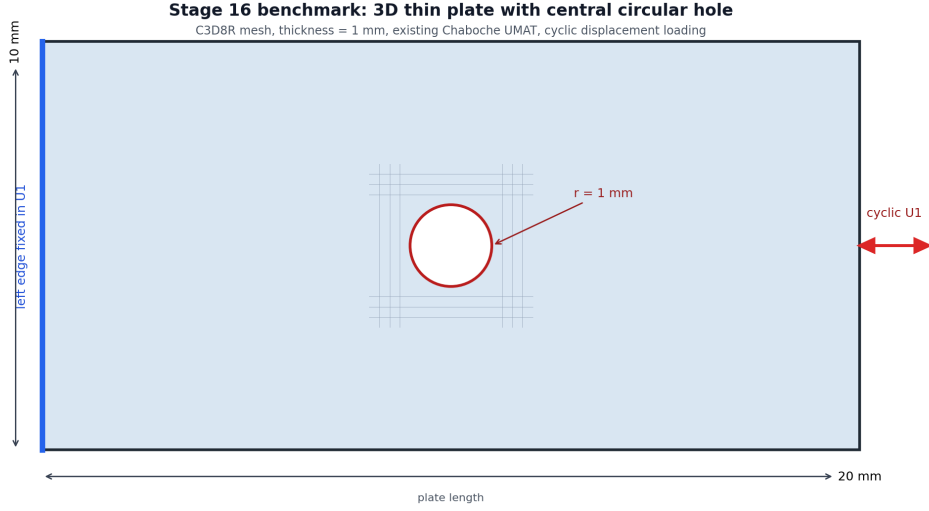


Figure 1: Plate-with-hole geometry used for the Stage 16N benchmark.

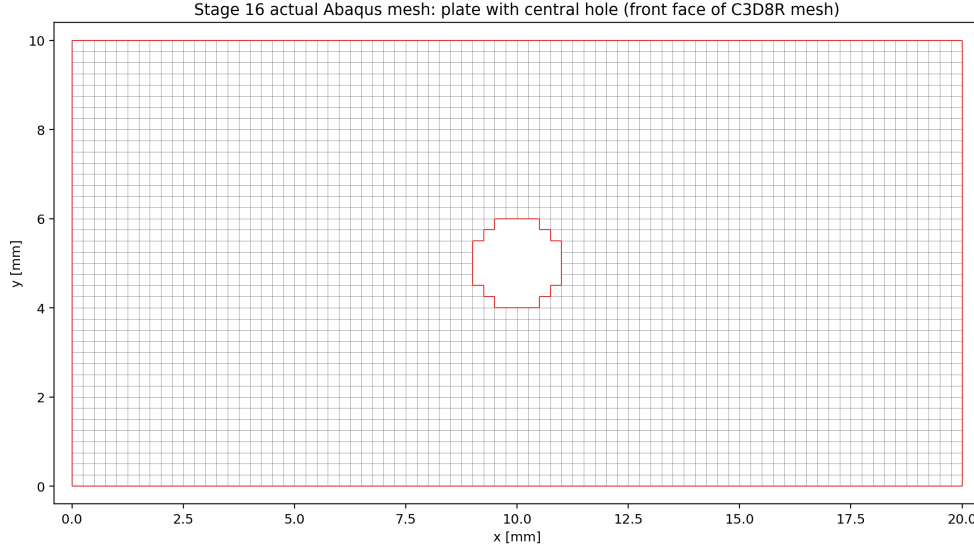


Figure 2: Actual structured C3D8 mesh with the element-wise approximation of the central hole.

## 2.1 Boundary Conditions and Loading

The left edge is fixed in the axial direction. Two anchor nodes constrain rigid-body modes. The right edge receives a cyclic axial displacement amplitude of  $\pm 0.10$ . The full reference contains 1000 full non-jump cycles. Selected field output is stored at cycles 1, 2, 10, 50, 100, 250, 500, 750, and 1000.

## 3 HPC and Production Policy

The first corrected 1000-cycle parallel reference used 30 OpenMP threads, but PBS accounting showed that the average effective CPU usage was far below the requested 30 cores. A 16-CPU verification run was therefore used to lock a more resource-efficient production policy.

Table 3: Locked Stage 16N production setting.

| Setting                        | Value                                                   |
|--------------------------------|---------------------------------------------------------|
| Production default             | 1 MPI rank $\times$ 16 OpenMP threads                   |
| PBS request                    | <code>select=1:ncpus=16:mpiprocs=1:ompthreads=16</code> |
| Abaqus launch                  | <code>cpus=16 mp_mode=threads</code>                    |
| Expected message-file line     | 1 MPI RANK $\times$ 16 THREADS                          |
| Maximum walltime per job       | 24:00:00                                                |
| Maximum simultaneous jobs      | 2                                                       |
| Maximum active Stage 16N cores | 32                                                      |

This policy is not claimed to perfectly saturate all 16 CPUs. It is a resource-efficient compromise that prevents the earlier serial fallback and avoids reserving 30 CPUs for a workload that did not use them efficiently.

### 3.1 Two-Job Scheduling Policy

From 2026-06-06 onward, Stage 16N solver planning assumes at most two simultaneous production jobs. Each job requests 16 CPU cores with one MPI rank and 16 OpenMP threads, so the intended active usage is 32 CPU cores. The two slots are used only after the relevant method gate has passed. First-time method checks, such as the first fixed cycle-jump case B1, run alone until extracted and compared. If B1 passes, B2 and B3 can be submitted together. If B1 fails badly, B2 and B3 remain held and the two slots are used instead for smaller or diagnostic B1 variants.

### 3.2 Mail Notification Policy

Future PBS jobs must explicitly request mail on abort, begin, and end:

```
#PBS -m abe
#PBS -M pr21vyci@mailserver.tu-freiberg.de
```

This was added because PBS accounting previously showed `Mail.Points = a`, which only requests abort/failure mail.

## 4 Full 1000-Cycle Non-Jump Reference

The full non-jump reference was completed using Abaqus/Standard with the corrected threaded launcher.

Table 4: Completed 1000-cycle reference run.

| Quantity            | Value                                                   |
|---------------------|---------------------------------------------------------|
| PBS job             | 1335408.mmaste02                                        |
| Abaqus job          | <code>stage16n_parallel_max_reference_1000cycles</code> |
| Walltime            | 17:56:43                                                |
| CPU time            | 178:03:42                                               |
| Requested resources | 30 CPUs, 135 GB                                         |
| Abaqus parallelism  | 1 MPI rank $\times$ 30 threads                          |
| Completed cycles    | 1000                                                    |
| Last partial cycle  | none                                                    |

Table 5: Reference evolution from cycle 1 to cycle 1000.

| Quantity            | Cycle 1  | Cycle 1000 | Change  |
|---------------------|----------|------------|---------|
| RF1_max             | 2356.834 | 2908.244   | +23.40% |
| RF1_min magnitude   | 2509.712 | 3074.823   | +22.52% |
| loop_area_abs       | 433.822  | 577.068    | +33.02% |
| HOLE_RING_MISES_MAX | 398.503  | 492.827    | +23.67% |
| HOLE_RING_SDV1_MAX  | 0.0561   | 26.0519    | —       |
| HOLE_RING_SDV8_MAX  | 0.2996   | 92.4455    | —       |
| HOLE_RING_SDV11_MAX | 7.9019   | 61.3076    | —       |

The evolution clearly exceeds the original continuation thresholds for global loop quantities and local hole-ring state variables. This confirms that the plate-with-hole benchmark is meaningful for cycle-jump validation.

## 5 Adaptive $\Delta N$ Table

The adaptive  $\Delta N$  table was computed from the completed 1000-cycle reference. It uses dense global cycle metrics and selected local field-output anchors. The controlling variable is the most sensitive monitored quantity, not only the global reaction force.

Table 6: Conservative adaptive  $\Delta N$  estimates from the 1000-cycle reference.

| Base cycle | Next anchor | Target | $\Delta N$ | Controlling variable |
|------------|-------------|--------|------------|----------------------|
| 1          | 2           | 2      | 1          | HOLE_RING_SDV8_MAX   |
| 2          | 10          | 3      | 1          | HOLE_RING_SDV8_MAX   |
| 10         | 50          | 14     | 4          | HOLE_RING_SDV8_MAX   |
| 50         | 100         | 61     | 11         | HOLE_RING_SDV1_MAX   |
| 100        | 250         | 124    | 24         | HOLE_RING_SDV1_MAX   |
| 250        | 500         | 299    | 49         | HOLE_RING_SDV1_MAX   |
| 500        | 750         | 575    | 75         | HOLE_RING_SDV1_MAX   |
| 750        | 1000        | 849    | 99         | HOLE_RING_MISES_MAX  |

The table shows that local state variables near the hole strongly restrict the allowable jump size. This supports the decision that any later fixed/adaptive jump validation must include local metrics, not only global RF loops.

### 5.1 Reinjection-Aware Controller Update

After the B0 initialization audit, HOLE\_RING\_SDV8\_MAX was downgraded from a hard pass/fail controller to a diagnostic-only variable because it already shows a large mismatch immediately after manual SDVINI/SIGINI initialization. A reinjection-aware table was therefore generated with SDV8 retained in diagnostic output but removed from the hard controller list.

Table 7: Reinjection-aware adaptive  $\Delta N$  estimates with SDV8 diagnostic-only.

| Base | Next anchor | Target | $\Delta N$ | Controller          |
|------|-------------|--------|------------|---------------------|
| 1    | 2           | 2      | 1          | HOLE_RING_SDV11_MAX |
| 2    | 10          | 3      | 1          | HOLE_RING_SDV1_MAX  |
| 10   | 50          | 15     | 5          | HOLE_RING_SDV1_MAX  |
| 50   | 100         | 61     | 11         | HOLE_RING_SDV1_MAX  |
| 100  | 250         | 124    | 24         | HOLE_RING_SDV1_MAX  |
| 250  | 500         | 299    | 49         | HOLE_RING_SDV1_MAX  |
| 500  | 750         | 575    | 75         | HOLE_RING_SDV1_MAX  |
| 750  | 1000        | 849    | 99         | HOLE_RING_MISES_MAX |

The first conservative fixed-jump gate remains cycle 100 to cycle 125, followed by normal continuation to cycle 250. This row is still controlled by HOLE\_RING\_SDV1\_MAX, not by diagnostic-only SDV8.

## 6 Stage 16N-B: State-Initialized Continuation Verification

Before any real cycle jump is allowed, the workflow must prove that stress and state-variable fields can be transferred into a new Abaqus run. The route used here is manual state-initialized continuation through SDVINI/SIGINI; it is not a native Abaqus restart. The originally intended verification cases were:

- B0-1: exact cycle 100 state, continue normally to cycle 250.
- B0-2: exact cycle 250 state, continue normally to cycle 500.
- B0-3: exact cycle 500 state, continue normally to cycle 1000.

B0-2 and B0-3 remain on hold. The B0-1 and initialization-audit results are now treated as the measured reinjection baseline rather than as proof of an exact local restart.

### 6.1 State Reader Implementation

The initial CSV preload design failed under threaded Abaqus because multiple threads attempted to load the same file concurrently. The state-transfer implementation was therefore changed to a direct-access binary reader. Each record contains:

S1, S2, S3, S4, S5, S6, SDV1, ..., SDV27

The record number is:

`record = (NOEL - 1) * 8 + NPT`

On the Abaqus/Intel Fortran environment used here, direct-access RECL is interpreted in 4-byte words. Therefore the correct value for 33 double-precision values is:

`RECL = 66`

## 7 B0-1 Cycle-100 to Cycle-250 State-Initialized Continuation

B0-1 completed the Abaqus solver successfully. PBS reported exit status 2 only because the original postprocessing path failed after solver completion; extraction was rerun manually afterward.

Table 8: B0-1 run accounting and solver status.

| Quantity           | Value                                   |
|--------------------|-----------------------------------------|
| PBS job            | 1336497.mmamaster02                     |
| Route              | cycle 100 state $\rightarrow$ cycle 250 |
| PBS walltime       | 01:41:36                                |
| PBS CPU time       | 10:52:42                                |
| Requested CPUs     | 16                                      |
| Abaqus solver      | completed                               |
| Abaqus parallelism | 1 MPI rank $\times$ 16 threads          |
| Scientific status  | Practical baseline accepted             |

Table 9: B0-1 comparison against the 1000-cycle reference at cycle 250.

| Quantity              | Relative error |
|-----------------------|----------------|
| RF1_max               | 0.259048%      |
| RF1_min               | 0.687538%      |
| loop_area_abs         | 0.967300%      |
| HOLE_RING_MISES_MAX   | 7.63648%       |
| HOLE_RING_S11_MAX_ABS | 9.58753%       |
| HOLE_RING_SDV1_MAX    | 0.360172%      |
| HOLE_RING_SDV8_MAX    | 10.9164%       |
| HOLE_RING_SDV11_MAX   | 7.27213%       |

The global response is excellent, but the local hole-ring stress and state-variable errors are too large for a clean exact-restart claim.

## 8 B0-1 Local Diagnostic Review

A pointwise hole-ring diagnostic was run to determine whether the high local error was only a maximum-location artifact. It was not. The reference and reinjection maxima for **SDV8** occurred at the same element and integration point.

Table 10: B0-1 SDV8 argmax diagnostic at cycle 250.

| Quantity                | Value              |
|-------------------------|--------------------|
| Reference SDV8 argmax   | element 1242, IP 7 |
| Reinjection SDV8 argmax | element 1242, IP 7 |
| Same argmax location    | true               |
| Reference value         | 91.1466598511      |
| Reinjection value       | 81.1967315674      |
| Argmax relative error   | 10.9164%           |



This made B0-1 a real local mismatch rather than a harmless maximum-location shift.

## 9 B0 Cycle-100 Initialization-Only Audit

The initialization audit was created to answer whether the local mismatch already exists immediately after exact cycle-100 initialization, before any cycle-100 to cycle-250 continuation.

Table 11: Cycle-100 binary reader and initialization audit.

| Quantity                             | Value                   |
|--------------------------------------|-------------------------|
| CSV-vs-binary maximum absolute error | $3.55 \times 10^{-15}$  |
| Common hole-ring element/IP records  | 480                     |
| SDV8 median relative error           | 12.7022%                |
| SDV8 95th percentile relative error  | 150.665%                |
| SDV8 maximum relative error          | 479.090%                |
| SDV8 argmax location                 | same element/IP, 1242/7 |

The binary extraction and reader pipeline is correct to machine precision. Therefore the local mismatch is not caused by the Python binary writer or direct-access file format. The mismatch appears during manual Abaqus initialization and the tiny equilibrium step.

### 9.1 Initialization Audit Interpretation

The most likely interpretation is that SDVINI/SIGINI inject stress and state variables correctly, but the manual state-initialized model does not reconstruct the compatible displacement, strain, solver-history, and full equilibrium state of the original reference analysis. Strain-like local state variables such as SDV8 are therefore adjusted during the first equilibrium solve. This means B0-1 is useful as a practical state-initialized continuation test, but it is not a mathematically exact Abaqus restart for local fields.

## 10 Reinjection Error Floor and Revised Metric Policy

The B0 audit closes Stage 16N-B with the following conclusion:

Global continuation works, binary state transfer works, but local exact restart is not achieved with SDVINI/SIGINI alone.

Future fixed and adaptive jump results must separate the manual reinjection error floor from the additional error caused by cycle-jump extrapolation. The primary pass/fail metrics are RF1 maximum/minimum, loop area, global hysteresis shape, HOLE\_RING\_SDV1\_MAX, HOLE\_RING\_MISES\_MAX, and HOLE\_RING\_S11\_MAX\_ABS. HOLE\_RING\_SDV8\_MAX remains important but is diagnostic-only because it is already affected by initialization before any cycle-jump approximation is applied.

## 11 Stage 16N-C Fixed State-Initialized Jump Scaffold

Stage 16N-C now begins with one conservative fixed jump:

Table 12: First fixed state-initialized jump case.

| Quantity                    | Value                              |
|-----------------------------|------------------------------------|
| Case name                   | B1_100_to_125_to_250               |
| Base state                  | cycle 100                          |
| Interpreted jump target     | cycle 125                          |
| Normal Abaqus continuation  | cycles 126–250                     |
| Comparison endpoint         | cycle 250                          |
| Initial state strategy      | zero-order hold from cycle 100     |
| Baseline for interpretation | B0-1 cycle 100 to 250 continuation |

The zero-order state strategy is intentionally conservative. It measures the error added by skipping cycles 101–125 on top of the known manual reinjection floor. If this case gives acceptable global error and controlled primary local errors, the next fixed jump cases can be expanded to cycle 250 to 300, cycle 500 to 575, and cycle 750 to 850.

### 11.1 Prepared-Only B2 and B3 Cases

B2 and B3 may be prepared while B1 is running, but their solver jobs are gated by the B1 result. They must not be submitted until B1 has been extracted, compared against the 1000-cycle reference, and interpreted against the B0 reinjection baseline.

Table 13: Prepared-only fixed-jump cases held behind the B1 gate.

| Case                 | Base state | Jump target | Compare endpoint |
|----------------------|------------|-------------|------------------|
| B2_250_to_300_to_500 | cycle 250  | cycle 300   | cycle 500        |
| B3_500_to_575_to_750 | cycle 500  | cycle 575   | cycle 750        |

If B1 passes, B2 and B3 are the first paired submission under the new two-job policy. If B1 fails, the paired slots should instead be used for B1-small and B1-diagnostic variants.

### 11.2 B1 Result

B1 completed successfully as an Abaqus job, but it did not pass the fixed-jump method gate. The comparison at cycle 250 gave good global errors but unacceptable primary local state errors.

Table 14: B1 fixed-jump comparison against the 1000-cycle reference and B0 baseline.

| Quantity              | B1 total error | B0 baseline | Additional error |
|-----------------------|----------------|-------------|------------------|
| RF1_max               | 0.361433%      | 0.259048%   | 0.102384%        |
| RF1_min               | 2.27018%       | 0.687538%   | 1.58264%         |
| loop_area_abs         | 1.26223%       | 0.967300%   | 0.294934%        |
| HOLE_RING_MISES_MAX   | 2.40141%       | 7.63648%    | -5.23507%        |
| HOLE_RING_S11_MAX_ABS | 1.21869%       | 9.58753%    | -8.36884%        |
| HOLE_RING_SDV1_MAX    | 10.2916%       | 0.360172%   | 9.93139%         |
| HOLE_RING_SDV8_MAX    | 16.1769%       | 10.9164%    | 5.26047%         |
| HOLE_RING_SDV11_MAX   | 11.2777%       | 7.27213%    | 4.00560%         |

The B1 status is **review**, controlled by HOLE\_RING\_SDV11\_MAX. Therefore B2 and B3 remain held. The two-job policy should next be used for smaller or diagnostic B1 variants rather than larger fixed jumps.

## 12 Failures and Fixes

Table 15: Important failures and fixes recorded in Stage 16N.

| Issue                             | Observation                                                                                                  | Fix / conclusion                                                                         |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Accidental serial-like run        | A 22-hour run reached only 592 full cycles.                                                                  | Parallel launcher corrected; later 1000-cycle reference completed.                       |
| 30-CPU over-allocation            | 30-thread run averaged about 10 effective cores.                                                             | 16 CPUs selected as production compromise.                                               |
| Missing start/end email           | PBS accounting showed <code>MailPoints = a</code> .                                                          | Added <code>#PBS -m a</code> and mail user to Stage 16N submit scripts.                  |
| CSV state-reader race             | Threaded Abaqus calls attempted concurrent CSV preload.                                                      | Replaced with direct-access binary state reader.                                         |
| Wrong direct-access record length | Abaqus/Intel runtime expected word-based RECL.                                                               | Set <code>RECL = 66</code> for 33 double values.                                         |
| Postprocessing path failure       | B0-1 solver completed but PBS exit status was 2.                                                             | Fixed extractor path and reran extraction manually.                                      |
| B0-1 local mismatch               | Global errors under 1%, local <code>SDV8</code> max error 10.9164%.                                          | Added pointwise diagnostic and initialization-only audit.                                |
| Initialization local mismatch     | Binary audit passed, but <code>SDV8</code> mismatch appeared immediately.                                    | Concluded manual <code>SDVINI/SIGINI</code> is not a mathematically exact local restart. |
| Metric policy mismatch            | Original adaptive table allowed <code>SDV8</code> to act as a hard controller.                               | Added reinjection-aware table with <code>SDV8</code> diagnostic-only.                    |
| B1 fixed-jump gate                | Global B1 errors were small, but <code>SDV1</code> and <code>SDV11</code> primary local errors exceeded 10%. | Hold B2/B3 and use the two-job policy for smaller/debug B1 variants.                     |

## 13 Current Conclusion

Stage 16N has successfully produced a full 1000-cycle non-jump reference and a conservative adaptive  $\Delta N$  table. The inhomogeneous plate-with-hole benchmark is scientifically useful because both global hysteresis quantities and local hole-ring state variables evolve significantly. The HPC workflow is now resource-controlled with a 16-CPU production policy and explicit start/end/failure mail notifications.

The important methodological limitation is exact local reinjection. B0-1 proves that the manual `SDVINI/SIGINI` workflow can run and reproduce the global force-displacement response well, but it does not reproduce local strain-like state variables exactly. The initialization-only audit shows that this local mismatch appears immediately during manual Abaqus initialization/equilibration, even though the binary reader is exact.

## 14 Next Work

The first conservative Stage 16N-C fixed state-initialized jump has now been run and compared. It reproduced global response fairly well, but primary local state errors were too large for a pass.

B0-2 and B0-3 remain on hold as exact-local-restart claims, and B2/B3 fixed jumps are also held until smaller/debug B1 variants clarify the local-state error.

1. Do not submit B2/B3 yet.
2. Prepare smaller/debug B1 variants, especially 100 to 112 to 250.
3. Use the second 16-core slot for an equilibration or cycle-phase diagnostic variant.
4. Keep SDV8 as a diagnostic warning metric, not a hard pass/fail metric.

## A Key Repository Evidence Files

- runs/chaboche\_umat/stage16\_abaqus\_inhomogeneous\_cycle\_jump\_benchmark/stage16n\_parallel\_ma
- runs/chaboche\_umat/stage16\_abaqus\_inhomogeneous\_cycle\_jump\_benchmark/stage16n\_adaptive\_de
- runs/chaboche\_umat/stage16\_abaqus\_inhomogeneous\_cycle\_jump\_benchmark/stage16n\_adaptive\_de
- runs/chaboche\_umat/stage16\_abaqus\_inhomogeneous\_cycle\_jump\_benchmark/STAGE16N\_REINJECTION
- runs/chaboche\_umat/stage16\_abaqus\_inhomogeneous\_cycle\_jump\_benchmark/STAGE16N\_FIXED\_JUMP\_
- runs/chaboche\_umat/stage16\_abaqus\_inhomogeneous\_cycle\_jump\_benchmark/stage16n\_exact\_reinj
- runs/chaboche\_umat/stage16\_abaqus\_inhomogeneous\_cycle\_jump\_benchmark/STAGE16N\_HPC\_MAIL\_PO